



ELEMENTARY MATH (CALCULUS)



HIGHER-ORDER DERIVATIVES: EXAMPLES & ITS APPLICATIONS

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LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- identify the different types of higher-order derivatives;
- calculate higher-order derivatives; and
- apply rules of differentiation for higher-orders.





INTRODUCTION

We're familiar with the first derivative, which tells us about the rate of change of a function. But what if we desire to understand how this rate of change itself changes, and then how that change is changing, and so forth? This is precisely what higher-order derivatives reveals.

In this session, we will explore the concept, significance, examples, and applications of higher-order derivatives.





FIRST-ORDER DERIVATIVE



What is the first derivative of $f(x) = x^4 + 2x^3 - 15x + 4$?

Solution:

Using the power rule which says when $f(x) = x^n$,

$$f' = nx^{n-1}$$

Then, the first-order derivative is

$$f'(x) = 4x^3 + 6x^2 - 15$$

NOTE: the derivative of a constant is zero i.e. $\frac{d(4)}{dx} = 0$.



FIRST DERIVATIVE

Differentiate $y = 3\tan x + 3$.

Solution:

Recall from the trigonometry rule, $\frac{d(\tan x)}{dx} = \sec^2 x$

Thus,

$$\frac{dy}{dx} = \frac{d(3\tan x)}{dx} + \frac{d(3)}{dx}$$

$$\frac{dy}{dx} = 3 \frac{d(\tan x)}{dx} + 0$$

$$\frac{dy}{dx} = 3 \sec^2 x$$



SECOND-ORDER DERIVATIVE

Given a function, $f(x) = \sqrt{x^3}$. Find the second derivative.

Solution:

It can also be written as $y = x^{\frac{3}{2}}$

The first derivative is $y' = \frac{3}{2}x^{\frac{1}{2}}$

The second derivative is gotten by differentiating y' again

$$y'' = \frac{3}{2} \cdot \frac{1}{2} x^{\frac{1}{2}-1}$$

$$y'' = \frac{3}{4} x^{-\frac{1}{2}}$$



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TASK

What is the second derivative of $y = \cos(3x)$?





THIRD-ORDER DERIVATIVE

What is the third derivative of $f(2) = \ln x$?

Solution:

Let $y = \ln x$

From the logarithmic rule: $\frac{d(\ln x)}{dx} = \frac{1}{x}$

The first derivative is $y' = \frac{1}{x}$ or x^{-1}

The second derivative is $y'' = -\frac{1}{x^2}$ or $-x^{-2}$

The third derivative is $y''' = \frac{2}{x^3}$ or $2x^{-3}$

Thus, $f'''(2) = \frac{2}{2^3} = \frac{2}{8} = \frac{1}{4}$



THIRD DERIVATIVE

Find the first three derivative of $\sin(5x) + e^{3x}$.

Solution:

$$\text{Let } \frac{dy}{dx} = \sin(5x) + e^{3x}$$

From the logarithmic rule: $\frac{d(\sin x)}{dx} = \cos x$ and $\frac{d(e^x)}{dx} = e^x$

The first derivative is $\frac{dy}{dx} = 5 \cos(5x) + 3e^{3x}$

The second derivative is $\frac{d^2y}{dx^2} = -25 \sin(5x) + 9e^{3x}$

The third derivative is $\frac{d^3y}{dx^3} = -125 \cos(5x) + 27e^{3x}$





FOURTH DERIVATIVE

Find the fourth derivative of $\ln(1 + x^2)$.

Solution:

Let $f(x) = \ln(1 + x^2)$.

Recall from chain rule, given $f(x) = g(f(x))$

If $y = f(u)$, and $u = g(x)$

Then $f'(x) = y'(u) \cdot u'(x)$

From the above question, $y = \ln(u)$ where $u = (1 + x^2)$

$$y'(u) = \frac{1}{u}, u'(x) = 2x$$

Therefore, the **first derivative** is $f'(x) = \frac{1}{u} \cdot 2x$

$$f'(x) = \frac{2x}{(1 + x^2)}$$





FOURTH DERIVATIVE

The first derivative, $f'(x) = \frac{2x}{(1+x^2)}$ is a quotient.

In order for us to get the second derivative, we will recall the quotient rule which states that, if $f(x) = \frac{u(x)}{v(x)}$

$$\text{then, } f'(x) = \frac{v(x)u'(x) - u(x)v'(x)}{(v(x))^2}$$

Thus, $u(x) = 2x, v(x) = 1 + x^2, u'(x) = 2, v'(x) = 2x$

$$f''(x) = \frac{(1 + x^2) \cdot 2 - 2x(2x)}{(1 + x^2)^2}$$

$$f''(x) = \frac{2 + 2x^2 - 4x^2}{(1 + x^2)(1 + x^2)}$$

$$f''(x) = \frac{2 - 2x^2}{1 + 2x^2 + x^4}$$





FOURTH DERIVATIVE

Therefore, the **second derivative** is $f''(x) = \frac{2-2x^2}{1+2x^2+x^4}$

Differentiating the second derivative, we get

$$u(x) = 2 - 2x^2, v(x) = 1 + 2x^2 + x^4$$

$$u'(x) = -4x, v'(x) = 4x + 4x^3$$

$$f'''(x) = \frac{(1 + 2x^2 + x^4)(-4x) - (2 - 2x^2)(4x + 4x^3)}{(1 + 2x^2 + x^4)^2}$$

$$f'''(x) = \frac{-4x - 8x^3 - 4x^5 - 8x - 8x^3 + 8x^3 + 8x^5}{(1 + 2x^2 + x^4)(1 + 2x^2 + x^4)}$$

$$f'''(x) = \frac{4x^5 - 8x^3 - 12x}{1 + 4x^2 + 6x^4 + 4x^6 + x^8}$$





FOURTH DERIVATIVE

Therefore, the **third derivative** is

$$f'''(x) = \frac{4x^5 - 8x^3 - 12x}{1 + 4x^2 + 6x^4 + 4x^6 + x^8}$$

Lastly, differentiating the third derivative, we obtain

$$u(x) = 4x^5 - 8x^3 - 12x, v(x) = 1 + 4x^2 + 6x^4 + 4x^6 + x^8$$

$$u'(x) = 20x^4 - 24x^2 - 12, v'(x) = 8x + 24x^3 + 24x^5 + 8x^7$$

$$f^{(4)}(x) = \frac{(1 + 4x^2 + 6x^4 + 4x^6 + x^8)(20x^4 - 24x^2 - 12) - (8x + 24x^3 + 24x^5 + 8x^7)(4x^5 - 8x^3 - 12x)}{(1 + 4x^2 + 6x^4 + 4x^6 + x^8)^2}$$

$$f^{(4)}(x) = \frac{-12x^{12} + 40x^{10} + 568x^8 + 416x^6 + 240x^4 + 24x^2 - 12}{(1 + 4x^2 + 6x^4 + 4x^6 + x^8)^2}$$





FOURTH DERIVATIVE

Therefore, the **fourth derivative** is

$$f^{(4)}(x) = \frac{-12x^{12} + 40x^{10} + 568x^8 + 416x^6 + 240x^4 + 24x^2 - 12}{(1 + 4x^2 + 6x^4 + 4x^6 + x^8)^2}$$

Note that the notation for the fourth derivative can be written as

$f^{(4)}(x)$ or $(f''''(x))'$.





TASK

Find the third derivative of $f(x) = xe^x$





APPLICATION TO THE REAL-WORLD

- Higher-order derivatives play a significant role in understanding motion. The second derivative represents acceleration which we often see in Physics.

For instance, when plotting the position of a car over time, the second derivative gives us the acceleration (how the car is speeding up or slowing down), and the third derivative tells us about jerk (how the acceleration is changing).

- In analyzing electrical circuits, control systems, vibrations in structures etc.

For example, in an alternating current (AC) circuit, the second derivative ($V''(t)$) may help determine how quickly the voltage is oscillating, aiding in the analysis and design of electrical systems.





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SUMMARY



SUMMARY

- Higher-order derivatives refer to derivatives of a function beyond the first order. The n th derivative of a function $f(x)$ is denoted by $f^{(n)}(x)$.
- They can be calculated using differentiation rules such as: the power rule, product rule, chain rule, etc., applied iteratively.
- Higher-order derivatives affect the smoothness and behaviour of functions - functions become smoother as the order of derivatives increases.
- They provide a deeper understanding of a function's behaviour, revealing information about acceleration, jerk, and other complex aspects.
- In physics, higher-order derivatives help model motion and forces. Engineers use them in signal processing and control systems. Economists analyze growth and utility functions with their aid.





FURTHER READING:

- "Precalculus" by James Stewart, Lothar Redlin, and Saleem Watson
- "Calculus" by James Stewart
- "Elementary Linear Algebra" by Howard Anton and Chris Rorres
- "Elementary Differential Equations and Boundary Value Problems" by William E. Boyce and Richard C. DiPrima

Online Resources:

MIT OpenCourseWare: Massachusetts Institute of Technology offers open access to various courses, including lectures and materials on functions. [MIT OpenCourseWare - Mathematics](#)



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THANK
YOU